

7.1: The general Binomial Expansion

1st Year Binomial Exp. Stated: $(a+b)^n = \sum_{r=0}^{n} \frac{n!}{(n-r)!\,r!} a^n b^{n-r}$, $n \in \mathbb{N}$

$\forall n \in \mathbb{R}$, $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots = \sum_{r=0}^{\infty} \frac{x^r}{r!}\left(\prod_{k=0}^{r-1}(n-k)\right)$

For $n \in \mathbb{N}$, the expansion is finite. $n \notin \mathbb{N} \Leftrightarrow$ expansion is infinite & only valid for $|x|<1$

For: $(a+x)^n = a^n\left(1+\frac{x}{a}\right)^n$, this is used as the expansion is infinite, & 1^n, $n \in \mathbb{R}$, $= 1$

7.2: Simplifying Algebraic Expressions

Adding fractions → find a common denominator
Multiplying fractions → Cancel in both denominator & numerator if they are in both.
Factorise numerator & denominator before cancellation. Watch out for difference of Two Squares!
State for which x the simplification is invalid, ie: den. $= 0$:

$\frac{(x+1)}{(x+1)} = 1$, $x \neq -1$

You can also divide through polynomials to find remainders, or add dummy numbers to help simplification:

① $\frac{x^2+2x-3}{x-2} = x+4+\frac{5}{x-2}$

② $\frac{x^2+2x-3}{x-2} = \frac{x^2}{x-2} + \frac{2x+4+1}{x-2} = \frac{x^2+1}{x-2} + 2 = \frac{(x^2-4)+5}{x-2} + 2$

$= x+4+\frac{5}{x-2}$

7.3: Partial Fractions

A proper algebraic fraction with a factorisable denominator can be decomposed into a sum of proper partial fractions:

$\frac{px+q}{(ax+b)(cx+d)} = \frac{A}{(ax+b)} + \frac{B}{(cx+d)}$ $\quad$ $\frac{px+q}{(ax+b)(cx+d)^2} = \frac{A}{(ax+b)} + \frac{B}{(cx+d)} + \frac{C}{(cx+d)^2}$

Substitution (sometimes rearrangement) can be used to find numerators once multiplied out.